Scenario 1: Identify Consistent High Performers

Question: Find employees who have consistently received a salary increase for the past 3 years.

Ans:

- Join the employee and salary table using employee_id.
- Use WHERE clause to apply condition salary increases for past 3 years.
- Use SELECT to display the employee name and salary.

Scenario 2: Customer Retention Analysis

Question: Find customers who made purchases in the last 6 months but not in the previous 6 months.

Ans:

- Join the customer and purchase table by using customer_id.
- Use WHERE to apply customers who made purchases in the last 6 months but not in the previous 6 months.

Scenario 3: Identify Products with Declining Sales

Question: Find products whose sales decreased by more than 30% compared to the previous month.

Ans:

- Calculate the sales percentage decreased.
- Then Apply Where clause to apply condition >0.30 percentage of sales.

Scenario 4: Calculate Moving Average of Sales

Question: Calculate a 3-month moving average for product sales.

Ans:

- Use a WINDOW FUNCTION to calculate the moving average using the AVG() function.
- Partition the data by product_id.

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- Order the data by sales_date.

Scenario 5: Detect Duplicate Records

Question: Find all duplicate records from a customer table based on email and phone number.

Ans:

- Use GROUP BY to aggregate the records based on email and phone number.
- Then use HAVING to get record which is more than one occurance.

Scenario 6: Identify Products with No Sales for 3 Consecutive Months

Question: List products that have not had any sales for 3 consecutive months.

Ans:

- Use Join to connect the product and sales table.
- Then use where clause with date function to get the products that have not had any sales for 3 consecutive months.

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